

Bayesian Learning

Lecture 9 - Hamiltonian Monte Carlo and Variational Inference

Mattias Villani

Department of Statistics
Stockholm University



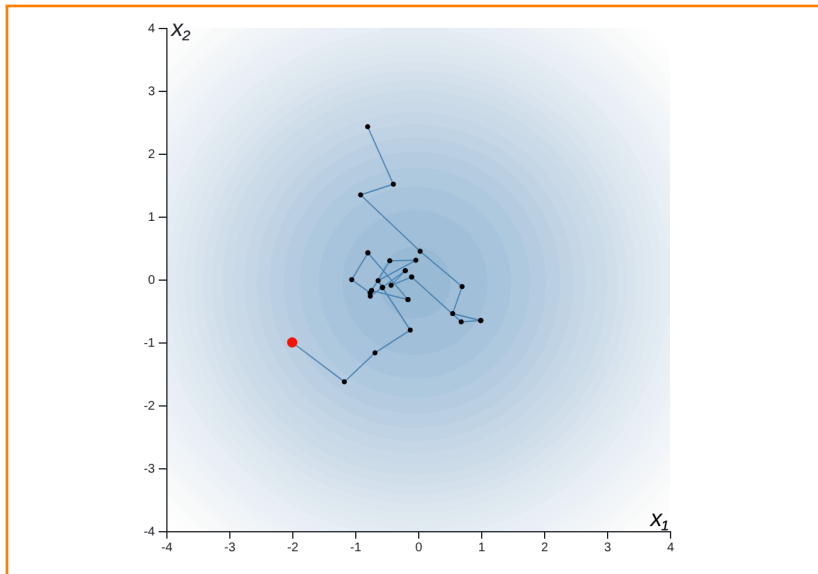
Lecture overview

- Hamiltonian Monte Carlo
- Variational Inference

Hamiltonian Monte Carlo

- When $\theta = (\theta_1, \dots, \theta_p)^\top$ is **high-dimensional**, $p(\theta|\mathbf{y})$ usually located in some subregion of $\mathbb{R}^p$ with complicated geometry.
- **Metropolis-Hastings (MH)**:
 - ▶ hard to find good proposal distribution $q(\cdot|\theta^{(i-1)})$.
 - ▶ RWM: must use small step size to avoid too many rejections.
- **Hamiltonian Monte Carlo (HMC)**:
 - ▶ distant proposals **and**
 - ▶ high acceptance probabilities.

Interactive - Random Walk Metropolis



HMC is based on ideas from physics

- Image a ball sliding over a friction-less surface: [illustration](#).
- **Location** of an object θ ($p \times 1$) (2D coordinates when $p = 2$).
- **Momentum** ϕ ($p \times 1$) - (mass $\times$ directional speed)
- **Hamiltonian** system
 - ▶ **Total energy** $H(\theta, \phi) = U(\theta) + K(\phi)$
 - ▶ U is the **potential energy** (“stored” energy, given by position)
 - ▶ K is the **kinetic energy** (energy from movement)
- **Hamiltonian Dynamics**: how the position and momentum changes over time:

$$\begin{aligned}\frac{d\theta}{dt} &= \frac{\partial H}{\partial \phi} \\ \frac{d\phi}{dt} &= -\frac{\partial H}{\partial \theta}\end{aligned}$$

Hamiltonian Monte Carlo

- HMC: MH algorithm to sample from posterior

$$p(\boldsymbol{\theta}|\mathbf{y}) \propto p(\mathbf{y}|\boldsymbol{\theta})p(\boldsymbol{\theta})$$

using **Hamiltonian dynamics to generate proposal $\boldsymbol{\theta}^*$** .

- Trick: add momentum parameters $\boldsymbol{\phi} = (\phi_1, \dots, \phi_p)^\top$ and sample from

$$p(\boldsymbol{\theta}, \boldsymbol{\phi}|\mathbf{y}) = p(\boldsymbol{\theta}|\mathbf{y})p(\boldsymbol{\phi})$$

- **Potential energy** is negative log posterior

$$U(\boldsymbol{\theta}) = -\log p(\boldsymbol{\theta}|\mathbf{y})$$

- **Momentum**: $\boldsymbol{\phi} \sim N(\mathbf{0}, \mathbf{M})$ where $\mathbf{M}$ is the mass matrix and

$$K(\boldsymbol{\phi}) = -\log p(\boldsymbol{\phi})$$

- If we could propose $\boldsymbol{\theta}$ in continuous time (spoiler: we can't), the MH acceptance probability would be one.

Approximate Dynamics via the Leapfrog algorithm

■ Hamiltonian Dynamics

$$\begin{aligned}\frac{d\theta}{dt} &= \mathbf{M}^{-1}\phi \\ \frac{d\phi}{dt} &= \frac{\partial \log p(\theta|\mathbf{y})}{\partial \theta}\end{aligned}$$

■ **Discretization** $\Rightarrow$ acceptance probability drops with ε .

Leapfrog algorithm

Function LEAPFROG($\theta^{(0)}, \phi^{(0)}, L, \epsilon, \mathbf{M}$)

for l in $1:L$ **do**

$$\tilde{\phi}^{(l)} = \phi^{(l-1)} + \frac{\epsilon}{2} \frac{\partial \log p(\mathbf{y}|\theta) p(\theta)}{\partial \theta} \Big|_{\theta^{(l-1)}}$$

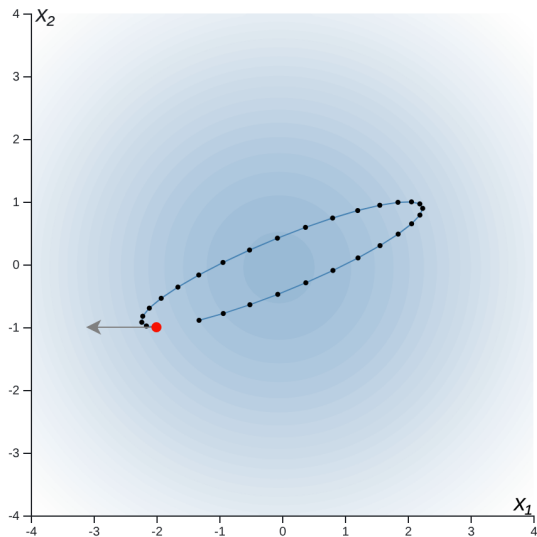
$$\theta^{(l)} = \theta^{(l-1)} + \epsilon \cdot \mathbf{M}^{-1} \tilde{\phi}^{(l)}$$

$$\phi^{(l)} = \tilde{\phi}^{(l)} + \frac{\epsilon}{2} \frac{\partial \log p(\mathbf{y}|\theta) p(\theta)}{\partial \theta_i} \Big|_{\theta^{(l)}}$$

end

Output: final position and momentum $\theta^{(L)}, \phi^{(L)}$.

Interactive - Leapfrog



The Hamiltonian Monte Carlo algorithm

for i in $1:(m+b)$ **do**

Draw initial momentum $\phi_s \sim N(0, \mathbf{M})$

Compute proposal with leaf-frog algorithm

$\theta^*, \phi^* = \text{LeapFrog}(\theta^{(i-1)}, \phi_s, L, \epsilon, \mathbf{M})$

Accept/reject proposal

Compute the acceptance probability

$$\alpha \leftarrow \min \left(1, \frac{p(\mathbf{y}|\theta^*)p(\theta^*)}{p(\mathbf{y}|\theta^{(i-1)})p(\theta^{(i-1)})} \frac{p(\phi^*)}{p(\phi_s)} \right)$$

Draw $u \sim \text{Uniform}(0, 1)$

if $u < \alpha$ **then**

$\theta^{(i)} = \theta^*$

else

$\theta^{(i)} = \theta^{(i-1)}$

end

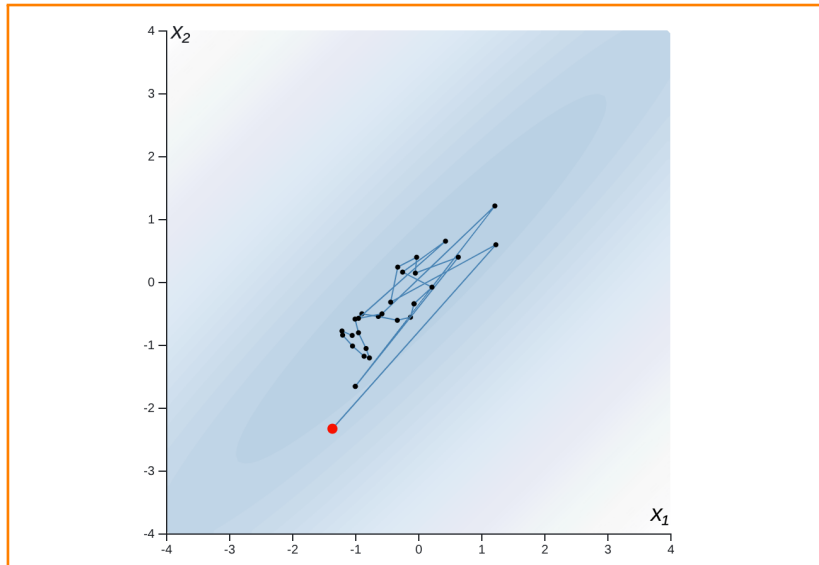
end

Output: draws $\theta^{(b+1)}, \dots, \theta^{(b+m)}$ from $p(\theta|\mathbf{y})$.

Tuning Hamiltonian Monte Carlo

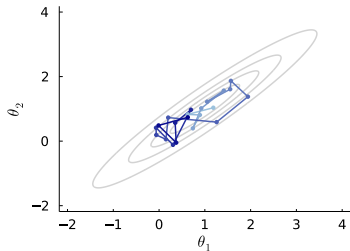
- HMC is very efficient, but **needs careful tuning** to work.
- **Tuning parameters:**
 - ▶ **stepsize** ε ,
 - ▶ **number of leapfrog** iterations L and
 - ▶ **mass matrix** M . (hello $J_x^{-1}(\tilde{\theta})$, our old friend)
- **No U-turn** sampler:
 - ▶ **Warm-up** to determine ε and L to get good acceptance rate.
 - ▶ Avoids U-turns in the Hamiltonian proposals.
- Drawbacks of HMC:
 - ▶ Need to **evaluate gradient of log posterior** many times during Hamiltonian iterations. Costly! (Subsampling HMC).

Interactive - HMC

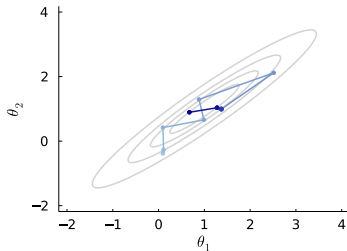


Comparing algorithms for bivariate normal

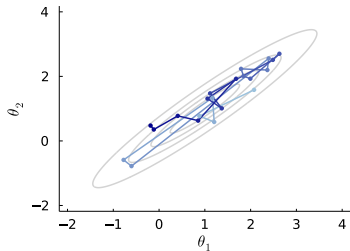
Gibbs



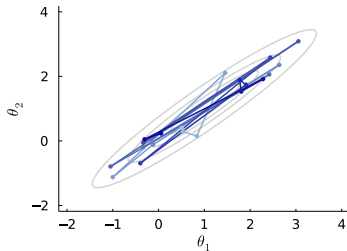
RWM - Identity M



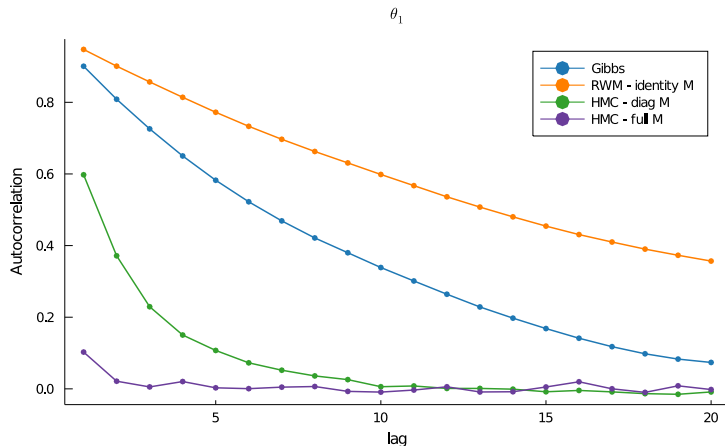
HMC - Diagonal M



HMC - Full M



Comparing algorithms for bivariate normal



Variational Inference

- Approximate posterior $p(\boldsymbol{\theta}|\mathbf{y})$ with (simpler) distribution $q(\boldsymbol{\theta})$.
- Before: **Normal approximation** from optimization:

$$q(\boldsymbol{\theta}) = N\left[\tilde{\boldsymbol{\theta}}, J_{\mathbf{x}}^{-1}(\tilde{\boldsymbol{\theta}})\right]$$

- **Mean field Variational Inference (VI)**:

$$q(\boldsymbol{\theta}) = \prod_{i=1}^p q_i(\theta_i)$$

- **Parametric VI**: Parametric family $q_{\lambda}(\boldsymbol{\theta})$ with parameters λ .
Example: $q(\boldsymbol{\theta}) = N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$. $\boldsymbol{\lambda} = (\boldsymbol{\mu}, \text{Chol}(\boldsymbol{\Sigma}))$.
- Find $q(\boldsymbol{\theta})$ that **minimizes the Kullback-Leibler divergence** between the true posterior p and the approximation q :

$$KL(q, p) = \int q(\boldsymbol{\theta}) \log \frac{q(\boldsymbol{\theta})}{p(\boldsymbol{\theta}|\mathbf{y})} d\boldsymbol{\theta} = E_q \left[\log \frac{q(\boldsymbol{\theta})}{p(\boldsymbol{\theta}|\mathbf{y})} \right].$$

Mean field approximation

- **Mean field VI** is based on factorized approximation:

$$q(\boldsymbol{\theta}) = \prod_{j=1}^p q_j(\theta_j)$$

- **No specific functional forms** are assumed for the $q_j(\theta_j)$.
- **Optimal densities** can be shown to satisfy:

$$q_j(\theta_j) \propto \exp \left(E_{-\theta_j} \ln p(\mathbf{y}, \boldsymbol{\theta}) \right)$$

where $E_{-\theta_j}(\cdot)$ is the expectation with respect to $\prod_{k \neq j} q_k(\theta_k)$.

- Note: no assumptions about parametric form of the $q_i(\theta)$.
- Optimal $q_i(\theta)$ often **turn out** to be parametric (normal etc).
- Just update hyperparameters in the optimal densities.
- **Structured mean field approximation**. Group subset of parameters in tractable blocks. Similar to Gibbs sampling.

Mean field approximation - Normal model

- **Model:** $X_i | \theta, \sigma^2 \stackrel{\text{iid}}{\sim} N(\theta, \sigma^2)$.
- **Prior:** $\theta \sim N(\mu_0, \tau_0^2)$ **independent** of $\sigma^2 \sim \text{Inv-}\chi^2(\nu_0, \sigma_0^2)$.
- **Mean-field approximation:** $q(\theta, \sigma^2) = q_\theta(\theta) \cdot q_{\sigma^2}(\sigma^2)$.
- Optimal densities

$$q_\theta^*(\theta) \propto \exp \left[E_{q(\sigma^2)} \ln p(\theta, \sigma^2, \mathbf{x}) \right]$$
$$q_{\sigma^2}^*(\sigma^2) \propto \exp \left[E_{q(\theta)} \ln p(\theta, \sigma^2, \mathbf{x}) \right]$$

Normal model - VB algorithm

■ Variational density for σ^2

$$\sigma^2 \sim \text{Inv} - \chi^2(\tilde{\nu}_n, \tilde{\sigma}_n^2)$$

$$\text{where } \tilde{\nu}_n = \nu_0 + n \text{ and } \tilde{\sigma}_n^2 = \frac{\nu_0 \sigma_0^2 + \sum_{i=1}^n (x_i - \tilde{\mu}_n)^2 + n \cdot \tilde{\tau}_n^2}{\nu_0 + n}$$

■ Variational density for θ

$$\theta \sim N(\tilde{\mu}_n, \tilde{\tau}_n^2)$$

where

$$\tilde{\tau}_n^2 = \frac{1}{\frac{n}{\tilde{\sigma}_n^2} + \frac{1}{\tau_0^2}}$$

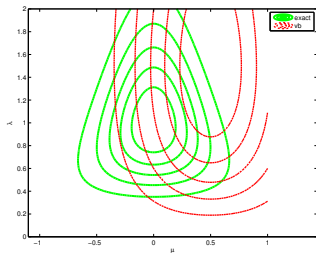
$$\tilde{\mu}_n = \tilde{w}\bar{x} + (1 - \tilde{w})\mu_0,$$

where

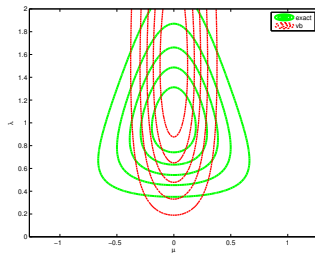
$$\tilde{w} = \frac{\frac{n}{\tilde{\sigma}_n^2}}{\frac{n}{\tilde{\sigma}_n^2} + \frac{1}{\tau_0^2}}$$

Normal example from Murphy ($\lambda = 1/\sigma^2$)

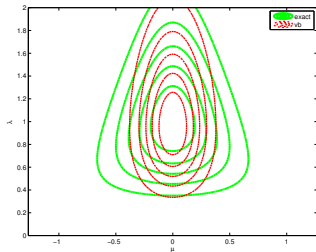
Initial values



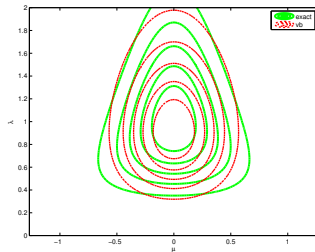
After updating q_μ



After updating q_{σ^2}



At convergence



Probit regression

- **Model:**

$$\Pr(y_i = 1 | \mathbf{x}_i) = \Phi(\mathbf{x}_i^T \boldsymbol{\beta})$$

- **Prior:** $\boldsymbol{\beta} \sim N(0, \Sigma_{\boldsymbol{\beta}})$. For example: $\Sigma_{\boldsymbol{\beta}} = \tau^2 I$.

- **Latent variable formulation** with $\mathbf{u} = (u_1, \dots, u_n)'$

$$\mathbf{u} | \boldsymbol{\beta} \sim N(\mathbf{X}\boldsymbol{\beta}, 1)$$

and

$$y_i = \begin{cases} 0 & \text{if } u_i \leq 0 \\ 1 & \text{if } u_i > 0 \end{cases}$$

- Factorized **variational approximation**

$$q(\mathbf{u}, \boldsymbol{\beta}) = q_{\mathbf{u}}(\mathbf{u}) q_{\boldsymbol{\beta}}(\boldsymbol{\beta})$$

VI for probit regression

■ VI posterior

$$\beta \sim N\left(\tilde{\mu}_\beta, \left(\mathbf{X}^\top \mathbf{X} + \Sigma_\beta^{-1}\right)^{-1}\right)$$

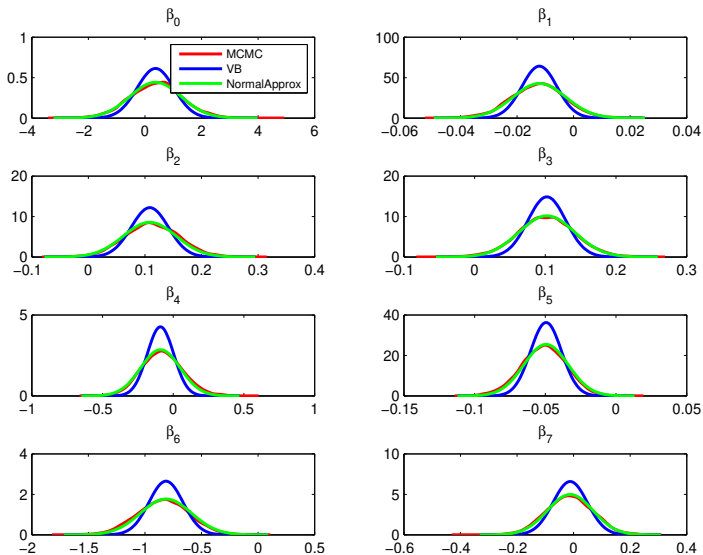
where

$$\tilde{\mu}_\beta = \left(\mathbf{X}^\top \mathbf{X} + \Sigma_\beta^{-1}\right)^{-1} \mathbf{X}^\top \tilde{\mu}_\mathbf{u}$$

and

$$\tilde{\mu}_\mathbf{u} = \mathbf{X} \tilde{\mu}_\beta + \frac{\phi(\mathbf{X} \tilde{\mu}_\beta)}{\Phi(\mathbf{X} \tilde{\mu}_\beta)^y [\Phi(\mathbf{X} \tilde{\mu}_\beta) - \mathbf{1}_n]^{1_n - y}}.$$

Probit example (n=200 observations)



Probit example

